running time for the `merge` operation on disjoint sets. This means that, even if there are millions or billions of elements, merging two subsets will, on average, only require at most 5 swaps.

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[1] A (no-SQL) key-value store used as a distributed object caching system.

[2] The set of all possible items – traditionally denoted as Universe (U) in set theory.

[3] Definition of protected visibility varies depending on the programming language: here, we assume a protected method or attribute is only visible by the class declaring it and its sub-classes. A private method, conversely is not visible by any classes inheriting from the one it is declared.

[4] This is a theory-intensive section

[5] As an implementation detail, we likely need to use referential equality here, when comparing lists.

[6] For a refresher on linked lists, or if you are not sure what `next` pointer is, check out appendix C.

[7] Remember that constant factors are irrelevant in *big-O* analysis, so `O(n/2) = O(n)`.

[8] Abbreviation for “if and only if”.

[9] A singleton is a set with exactly one element.

[10] This section is theory-intensive and features advanced concepts.

[11] Check out appendix C, section C.1.3.

[12] This section is theory-intensive and features advanced concepts.

[13] It is possible to find plenty of literature on the subject. Be aware, though: it's a very interesting reading, but challenging.

[14] For an introduction to graphs, see appendix G.

[15] Unsupervised machine learning deals with making sense of "unlabeled" data (i.e. data that has not been classified or categorized), with the goal of finding a structure in the raw data.

[16] A dendrogram (from Greek dendro "tree" and gramma "drawing") is a tree diagram specifically used to illustrate the arrangement of the clusters produced by hierarchical clustering.